EV Market Performance & Insights

Project Documentation: A Tableau Data Visualisation and Analytics Project

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Tool: Tableau Public | Dataset: ev_market_test.csv and ev_market_train.csv

Project links: Dashboard (Tableau Public), Story (Tableau Public), Web application (https://ev-market-performance-3.onrender.com/)

Abstract

Electric vehicles (EVs) are one of the fastest-changing segments of the global automotive industry. Buyers, manufacturers, investors and policy makers all need to understand how EVs differ in range, battery technology, charging speed, efficiency, price and sales performance. This project analyses an EV market dataset of 100,000 vehicle records and 37 fields, spanning 200 manufacturers, five battery chemistries, three drivetrain types and model years 2018 to 2026.

The data was loaded into Tableau Public, where nine worksheets were built: Battery Capacity by Battery Type, Charging Time by Drivetrain, EV Range Analysis, EV Price Analysis, Efficiency by Drivetrain, Battery Capacity vs Range, Manufacturing Country map, Annual EV Sales Trend, and Sales by Manufacturer. These were combined into one interactive dashboard and a narrated Tableau Story, and the findings were also published through a web application. This document describes the project step by step: the problem, the data, preparation, visual design, dashboard, findings, conclusion and future scope.

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1. Introduction and Problem Statement

1.1 Background

The global shift from combustion engines to electric mobility has produced a large and diverse EV market. Dozens of manufacturers now offer vehicles with different battery chemistries (LFP, NMC, NCA, Sodium-Ion and Solid-State), different drivetrains (AWD, FWD and RWD), and very different ranges, prices and charging speeds. For a decision maker, a raw spreadsheet of thousands of rows does not make these trade-offs visible.

1.2 Problem Statement

The market has too many models, specifications and manufacturers to compare by reading tables. Without a clear visual summary it is hard to answer questions such as:

- Which battery type delivers the largest capacity and which sells the most?
- Does a larger battery always give more range?
- Which drivetrain charges fastest and which is most efficient?
- Which manufacturers lead in range, and where are vehicles manufactured?
- How have sales changed from 2018 to 2026?

This project solves the problem by converting the raw dataset into an interactive visual analytics solution that answers these questions at a glance.

1.3 Motivation

Good visual analytics shortens the time between data and decision. A dashboard that lets a user filter by manufacturer or battery type gives the same insight in seconds that would otherwise take hours of manual spreadsheet work.

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2. Objectives and Scope

2.1 Objectives

1. Collect and load the EV market data into Tableau Public.
2. Understand the structure of the data and classify each field as a dimension or measure.
3. Prepare the data so that it is reliable for analysis.
4. Build clear worksheets covering battery, range, charging, efficiency, price, geography and sales trend.
5. Combine the worksheets into an interactive dashboard.
6. Present the findings in a Tableau Story with a clear narrative.
7. Publish the results online and document the full workflow.

2.2 Scope

The project covers descriptive and comparative analysis of EV specifications and sales across manufacturers, battery types, drivetrains, countries and model years. It does not include machine-learning forecasting, although the presence of a train file and a test file leaves room for this in future work.

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3. Tools and Technologies

Tool	Purpose
Tableau Public	Data connection, worksheet creation, dashboard and story
CSV files	Source data format (ev_market_test.csv, ev_market_train.csv)

Tool	Purpose
Tableau Maps (Mapbox / OpenStreetMap)	Geographic view of manufacturing countries
Render	Hosting of the web application
Windows Snipping Tool	Capturing screenshots used in this documentation

Tableau Public was chosen because it is free, supports interactive dashboards, and lets work be published online and shared by link.

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4. Data Collection

4.1 Source

The project uses an EV market dataset supplied as two CSV files:

- **ev_market_train.csv** – the training portion of the data.
- **ev_market_test.csv** – the test portion, which is the file connected for the visualisations.

4.2 Connecting the Data

The workbook was created in Tableau Public and the text-file connector was used to open the folder containing the CSV files. The Data Source tab shows the connection named *ev_market_test*, with the file ev_market_test.csv placed on the canvas. Tableau reported **37 fields and 100,000 rows** for this file. The optional Data Interpreter was left unticked because the CSV already had a clean header row and consistent columns.

4.3 Why a CSV Connection

A CSV extract is fast, portable and works with Tableau Public. Because the whole table is held in a single file, no joins were needed, which keeps the model simple and avoids duplication of rows.

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5. Data Understanding

5.1 Dataset Size and Coverage

Item	Value
Records (rows)	100,000
Fields (columns)	37
Manufacturers	200
Battery types	5 (LFP, NCA, NMC, Sodium-Ion, Solid-State)
Drivetrains	3 (AWD, FWD, RWD)
Model years	2018 to 2026
Manufacturing countries	10

5.2 Fields Used

The fields appear in Tableau's Data pane, grouped automatically into dimensions (text and geographic fields) and measures (numeric fields).

Dimensions

- Vehicle Id, Model Name, Manufacturer
- Manufacturing Country (assigned a geographic role)
- Market Region
- Battery Type, Category, Drivetrain
- Recommended Use Case

Measures

- Model Year, Annual Sales Units, Price Usd, Range Km
- Battery Capacity Kwh, Battery Cycle Life, Battery Degradation Pct
- Charging Ac Time Hr, Charging Dc Fast Time Min, Max Dc Charging Kw
- Efficiency Wh Per Km, Acceleration 0 100 Sec
- Co2 Saved Tons Per Year, Resale Value Pct 3Yr
- Cargo Capacity Liters, Infotainment Screen Inch
- Insurance Cost Usd Yearly, Customer Rating, Autonomous Level

5.3 Sample Records

The first rows of the data preview show the structure. For example, vehicle EV412673 is a Zenith-365 made by VoltRides in 2018 in India for the Latin America market, while EV372491 is a Nimbus-565 by IonRides made in South Korea in 2026 for the Europe market. This confirms that every row is a single vehicle model record with its manufacturer, year, country and market region.

5.4 Observation on the Nature of the Data

Several patterns in the data, such as evenly distributed manufacturers (about 500 records each) and stepped, block-like patterns in the capacity-versus-range scatter plot, suggest that the dataset is a structured, possibly synthetically generated, market dataset. The results should therefore be read as illustrative of how to analyse an EV market, not as official industry statistics.

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6. Data Cleaning and Preparation

Data preparation was carried out before any visual was built, so that every chart rests on reliable inputs.

6.1 Checks Performed

Step	What was done	Why it matters
Header and structure check	Confirmed the first row contains field names and all 37 columns load correctly	Prevents misaligned columns
Data type verification	Confirmed text fields show the *Abc* type and numeric fields show the number (#) type in the Data Source tab	Ensures correct aggregation (SUM, AVG)
Geographic role	Manufacturing Country was given a geographic role (shown by the globe icon)	Allows Tableau to generate Latitude and Longitude for the map
Year handling	Model Year was kept as a numeric dimension on the axis	Gives a clean 2018 to 2026 time axis

Step	What was done	Why it matters
Naming	Field names were checked for readability (for example, Battery Capacity Kwh, Range Km)	Gives clear axis labels
Duplicate and null review	Vehicle Id was used as the unique key and key numeric fields were reviewed for missing values	Avoids double counting and distorted averages
Train/test handling	The test file was used for the dashboard; the train file is kept for future modelling	Keeps the analysis and any later model separate

6.2 Dimension and Measure Decisions

A key decision is whether a numeric field should be summed or averaged. Totals make sense for volume measures (Annual Sales Units), but specification fields such as Battery Capacity, Charging Time and Efficiency are properties of a vehicle, so they were aggregated with **average (AVG)**. Range Km in the manufacturer chart is shown as a **sum** to compare each manufacturer's combined range output, and as an **average** in the scatter plot to compare individual capacity and range.

6.3 Result of Preparation

After these steps the data was ready for analysis: 100,000 clean rows, correctly typed fields, a geographic field for mapping and no need for joins or unions.

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7. Analysis Approach and Worksheet Design

7.1 Approach

The analysis follows five themes, each answering a decision question.

Theme	Question	Worksheet
Battery	Which battery type offers the most capacity?	(Battery Capacity Kwh)
Charging	Which drivetrain charges fastest?	Charging Time by Drivetrain

Theme	Question	Worksheet
Range and efficiency	Who leads in range, and how efficient are the vehicles?	EV Range Analysis, Efficiency by Drivetrain, Battery Capacity vs Range
Price	How are vehicles priced?	EV Price Analysis
Market	Where are vehicles made and how do sales trend?	Manufacturing Country, Annual EV Sales Trend

7.2 Chart Selection Principles

- **Bar charts** for comparing a measure across a small number of categories (battery type, drivetrain).
- **Highlight table (heat map)** for comparing a measure across two dimensions (manufacturer and drivetrain).
- **Scatter plot** for the relationship between two numeric measures (capacity and range).
- **Symbol map** for geographic distribution.
- **Line chart** for change over time.

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8. Worksheet-by-Worksheet Analysis

8.1 Battery Capacity by Battery Type

Build: Battery Type placed on Columns and AVG(Battery Capacity Kwh) on Rows, producing a bar chart with five bars: LFP, NCA, NMC, Sodium-Ion and Solid-State.

Reading the chart: All five chemistries have very similar average capacity, at roughly 78 kWh. The Solid-State bar is marginally the tallest and NMC marginally the shortest. The Tableau status bar shows the sum of the five averages as 391.7, which corresponds to an overall average of about 78.3 kWh.

Insight: Battery chemistry in this dataset does not determine pack size. Capacity is a design choice made per vehicle, while the chemistry affects other factors such as cost, cycle life and safety.

8.2 Charging Time by Drivetrain

Build: Drivetrain on Columns and AVG(Charging Ac Time Hr) on Rows.

Reading the chart: All-wheel drive (AWD) vehicles have the longest average AC charging time, at about 13 hours. Front-wheel drive (FWD) and rear-wheel drive (RWD) are close to each other at about 10.5 hours.

Insight: AWD vehicles take about 2.5 hours longer to charge on AC than FWD or RWD. A plausible reason is that AWD vehicles tend to carry larger packs and more drive hardware, though the dataset alone cannot prove cause.

8.3 EV Range Analysis

Build: Manufacturer on Columns and SUM(Range Km) on Rows, with the Bar mark type. The worksheet contains 200 marks, one per manufacturer, and the total range across all manufacturers is 32,430,696 km.

Reading the chart: Bars range roughly between 140K and 180K km. Hovering over a bar shows the exact value; for example, AxleDynamics shows a combined range of 174,559 km. CircuitWorks, AxleEV, AxleAutomotive, DriveWorks and several others sit near the top, while manufacturers such as AmpereMobility and CurrentRides sit at the lower end.

Insight: Performance is tightly clustered. The spread between the best and weakest manufacturer is small compared with the total, so range alone is not a strong differentiator, and buyers should compare manufacturers on a combination of factors.

8.4 EV Price Analysis

Build: Vehicle Id on Rows and Price Usd on Columns, giving a horizontal bar per vehicle. The axis runs from 0 to 400K USD.

Reading the chart: The vehicles shown on the dashboard (for example EV200001, EV200009, EV200011) have very different prices, ranging from very low values to well above 100K USD.

Insight: Price varies widely even between consecutive records, so price is driven by model specification rather than by manufacturer alone. The chart is scrollable so that individual vehicles can be inspected.

8.5 Efficiency by Drivetrain

Build: Manufacturer on Rows, Drivetrain on Columns, and AVG(Efficiency Wh Per Km) on both Colour and Text, using the Square mark type. The sheet has 200 rows by 3 columns, giving 600 marks.

Reading the chart: The colour legend runs from 186.9 to 342.8 Wh/km. Darker cells indicate higher energy consumption per kilometre, and therefore lower efficiency. For example, AmpereEV with FWD is among the darkest cells at 323.4 Wh/km, while AxleAuto with AWD is among the lightest at 189.1 Wh/km.

Insight: Efficiency depends on the manufacturer and drivetrain combination. No single drivetrain is the most efficient across all manufacturers, so the heat map is useful for identifying specific best and worst combinations.

8.6 Battery Capacity vs Range

Build: AVG(Battery Capacity Kwh) on Columns and AVG(Range Km) on Rows, with Manufacturer on Colour and Vehicle Id and Model Name on Detail. The Circle mark type is used, giving 100,000 marks.

Reading the chart: Capacity runs from near 0 to about 400 kWh and range from about 50 to 650 km. There is a clear positive relationship in the lower and middle section: as capacity rises from about 20 to 150 kWh, range rises from about 50 to 650 km. Beyond about 150 kWh the points form a wide plateau between roughly 140 and 360 km, meaning very large batteries do not give proportionally more range.

Insight: There are diminishing returns from adding battery capacity. Efficient design matters more than simply adding larger batteries.

8.7 Manufacturing Country

Build: Manufacturing Country (geographic role) on Detail, which generates Longitude and Latitude. SUM(Annual Sales Units) is placed on both Colour and Size, giving 10 marks.

Reading the chart: Ten manufacturing countries appear on the world map, including the USA, European countries such as Germany, France and Sweden, and Asian countries such as India, China, Japan and South Korea. Sales by country are between 75,569,268 and 78,870,003 units, and the grand total is 772,608,124 units.

Insight: The manufacturing base is globally spread and evenly balanced. The gap between the highest and lowest country is only about 3.3 million units, or around 4 percent.

8.8 Annual EV Sales Trend

Build: Model Year on Columns, SUM(Annual Sales Units) on Rows, Battery Type on Colour, and a line mark. The sheet contains 45 marks (five battery types over nine years).

Reading the chart:

Battery type	Approx. yearly sales	Pattern
NMC	27M to 30M	Highest in most years, slight decline after 2021
LFP	26M to 29M	Dip in 2020, then recovery and stability
NCA	13M to 16M	Volatile, low point in 2025
Sodium-Ion	about 8M to 9M	Very stable
Solid-State	5M to 7.5M	Peak in 2023, then lower

Insight: NMC and LFP dominate volume, and the two converge by 2025 and 2026. NCA holds a middle position, while the newer Sodium-Ion and Solid-State chemistries hold smaller but meaningful shares.

8.9 Sales by Manufacturer

Build: Manufacturer on Columns and SUM(Annual Sales Units) on Rows, shown on the dashboard.

Reading the chart: Each of the 200 manufacturers sells in a similar band of roughly 3M to 4M units, so no single manufacturer dominates. The chart is used mainly for orientation and for filtering the other charts.

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9. Dashboard Design

9.1 Purpose

The dashboard, titled **Dashboard 1**, brings all the worksheets into one view so that a user can compare battery, range, charging, efficiency, price, geography and sales without switching sheets.

9.2 Layout

Position	Content
Top left	Sales by Manufacturer
Top centre	Charging Time by Drivetrain

Position	Content
Middle left	EV Price Analysis
Middle centre	Battery Capacity vs Range
Lower left	EV Range Analysis
Lower centre	(Battery Capacity Kwh) bar chart
Bottom left	Efficiency by Drivetrain heat map
Bottom centre	Annual EV Sales Trend
Right column	Efficiency colour legend, Manufacturer legend, Manufacturing Country map, Annual Sales Units legends, Battery Type legend

9.3 Design Choices

- **Consistent colour:** a single blue tone is used for simple bar charts, so that colour only carries meaning where it is needed (battery type in the trend chart, manufacturer in the scatter plot).
- **Reading order:** the layout runs left to right and top to bottom, from market overview (sales) to technical detail (range, efficiency) and finally to trend and geography.
- **Titles:** every chart has a clear title so that it can be understood on its own.
- **Legends grouped on the right:** keeps the working area free for the charts.

9.4 Interactivity

The dashboard supports hovering for tooltips (for example the exact range of a single manufacturer), scrolling on the price and heat map charts, and map zoom. Highlighting and filter actions allow a user to select a battery type or manufacturer and see the related marks respond across sheets.

9.5 Link

The published dashboard is available on Tableau Public at:
https://public.tableau.com/app/profile/manika.gupta/viz/EVMarketPerformanceInsights_17935594784180/Dashboard1

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10. Story Design

10.1 Purpose

A dashboard is a tool for exploration, while a story is a guided narrative. The Tableau Story (**Story 1**) walks a viewer through the findings in a deliberate order, each point carrying a caption that explains what to notice.

10.2 Story Flow

Story point	Content	Message
1	Market overview and sales by manufacturer	The market is broad and balanced
2	Battery capacity by battery type	Chemistry does not change pack size
3	Capacity vs range	More capacity helps range up to a point
4	Charging time and efficiency by drivetrain	Drivetrain affects charging and energy use
5	Manufacturing country map	Production is spread across ten countries
6	Annual sales trend	NMC and LFP lead, others fill niches
7	Summary of findings	Decision-ready conclusions

10.3 Link

The Story is published at:
https://public.tableau.com/app/profile/manika.gupta/viz/EVMarketPerformanceInsightsStory/Story1?publish=yes

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11. Web Application Deployment

11.1 Purpose

To make the insights available to people who do not use Tableau, the project is also published as a web application at https://ev-market-performance-3.onrender.com/.

11.2 Hosting

The app is hosted on Render, a cloud platform that deploys directly from a code repository and provides a public URL. Free-tier services on Render may take a short time to wake up on the first visit after a period of inactivity, so the first load can be slow.

11.3 Relationship to the Tableau Work

The Tableau dashboard is the analytical core, and the web application presents the same insights through a browser-friendly interface. Both use the same dataset and cover the same questions, so findings remain consistent.

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12. Key Findings and Results

#	Finding	Evidence
1	All battery types have a similar average capacity of about 78 kWh	Battery capacity bar chart
2	AWD vehicles take longest to charge on AC, about 13 hours against about 10.5 hours for FWD and RWD	Charging time chart
3	Manufacturer range is tightly grouped between roughly 140K and 180K combined km	EV range analysis
4	Efficiency ranges from 186.9 to 342.8 Wh/km, and the best drivetrain differs by manufacturer	Efficiency heat map
5	Range grows with capacity up to roughly 150 kWh and then flattens	Capacity vs range scatter
6	Ten manufacturing countries each account for 75.6M to 78.9M sales units, out of a total of 772.6M	Manufacturing country map

#	Finding	Evidence
7	NMC and LFP lead sales volumes, NCA is mid-sized, and Sodium-Ion and Solid-State are smaller	Annual sales trend
8	Sales are stable between 2018 and 2026, with no chemistry showing explosive growth	Annual sales trend

12.1 Overall Interpretation

The market in this dataset is mature and balanced. There is no dominant manufacturer, country or battery chemistry that is far ahead of the rest in specifications, and differences come from detailed trade-offs, such as drivetrain against charging time, or capacity against range.

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13. Business Recommendations

For manufacturers

- Focus on efficiency (Wh/km) rather than simply larger batteries, because very large packs give diminishing returns in range.
- Investigate why AWD models charge more slowly and consider charging hardware improvements for them.

For buyers and fleet operators

- Compare vehicles on range per kWh and efficiency, not on battery size alone.
- Choose FWD or RWD where AC charging time is a priority.

For investors and planners

- NMC and LFP are the volume leaders, so supply chains for these chemistries are the most important in the near term.
- Sodium-Ion and Solid-State are smaller but stable, and are worth monitoring as costs and technology change.

For policy makers

- Manufacturing is distributed across ten countries, so supply-chain risk is spread rather than concentrated.

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14. Limitations

- The data appears structured or synthetic, so findings may not match real-world market statistics.
- Only descriptive analysis is performed; no statistical tests or forecasting models are included.
- Averages hide variation within each group, such as differences between individual models of the same manufacturer.
- Charts with 200 manufacturers are dense and require scrolling or filtering to read individual bars.
- The dashboard is limited by Tableau Public's data size and privacy rules, as all published data is public.

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15. Conclusion and Future Scope

15.1 Conclusion

This project took a raw EV market dataset of 100,000 records and turned it into a set of connected visual tools: nine worksheets, an interactive dashboard, a guided story and a hosted web application. Working step by step, the problem was defined, the data was collected and loaded, prepared and typed correctly, visualised with chart types matched to each question, and summarised as findings.

The main conclusion is that the EV market in this dataset is balanced, and that choices about drivetrain, efficiency and battery size matter more than the choice of battery chemistry or manufacturer alone. The project shows how Tableau can make a large dataset understandable to both technical and non-technical audiences.

15.2 Future Scope

1. **Predictive modelling:** use the training file to build models predicting price, range or sales, and validate them on the test file.
2. **More filters and parameters:** add filters for market region, model year and category, and parameters for what-if comparison.
3. **Calculated fields:** create metrics such as range per kWh, price per km of range and cost of ownership.

4. **Real market data:** repeat the analysis with official sales and specification data for validation.
5. **Customer insight:** analyse customer rating, resale value and battery degradation to understand ownership experience.
6. **Sustainability analysis:** use CO2 saved per year to quantify the environmental benefit of each segment.
7. **Automated refresh:** connect to a live data source so that the dashboard updates automatically.

Appendix A: Worksheet Summary

Sheet	Columns	Rows	Mark	Marks
(Battery Capacity Kwh)	Battery Type	AVG(Battery Capacity Kwh)	Bar	5
Charging Time by Drivetrain	Drivetrain	AVG(Charging Ac Time Hr)	Bar	3
EV Range Analysis	Manufacturer	SUM(Range Km)	Bar	200
EV Price Analysis	Price Usd	Vehicle Id	Bar	per vehicle
Efficiency by Drivetrain	Drivetrain	Manufacturer	Square	600
Battery Capacity vs Range	AVG(Battery Capacity Kwh)	AVG(Range Km)	Circle	100,000
Manufacturing Country	Longitude	Latitude	Map symbol	10
Annual EV Sales Trend	Model Year	SUM(Annual Sales Units)	Line	45

Appendix B: Screenshots to Insert

Insert the following screenshots under the matching sections: Data Source tab (Section 4), each worksheet (Section 8), the Dashboard (Section 9) and the Story (Section 10).

Appendix C: Project Links

- Dashboard: https://public.tableau.com/app/profile/manika.gupta/viz/EVMarketPerformanceInsights_17935594784180/Dashboard1
- Story: https://public.tableau.com/app/profile/manika.gupta/viz/EVMarketPerformanceInsightsStory/Story1?publish=yes
- Web app: https://ev-market-performance-3.onrender.com/